

## 27 Supersymmetric Gauge Theories

### 27.6 Perturbative Non-Renormalization Theorems

From the beginning, several of the ultraviolet divergences in ordinary renormalizable quantum field theories were found to be absent in the supersymmetric versions of these theories. With the development in 1975 of supergraph techniques, in which all particles in each supermultiplet are considered together, it became possible to show that some radiative corrections are not only finite, but are absent altogether in perturbation theory [5]. Supergraphs will be described in detail in Chapter 30, but as it happens they are not needed to prove the most important non-renormalization theorems. This section will give a version of a method developed by Seiberg [6] in 1993, which showed how the non-renormalization theorems may be easily obtained from simple considerations of symmetry and analyticity.

Consider a general renormalizable supersymmetric gauge theory with a number of left-chiral superfields  $\Phi_n$  and/or gauge superfields  $V_A$ . As mentioned in Section 27.3, if we remove the factor  $g$  in the  $t_A$  and  $C_{ABC}$  and include it instead in the gauge superfields, then the Lagrangian density will be of the form

$$\mathcal{L} = \left[ \Phi^\dagger e^{-V} \Phi \right]_D + 2 \operatorname{Re} [f(\Phi)]_{\mathcal{F}} + \frac{1}{2g^2} \operatorname{Re} \left[ \sum_{A\alpha\beta} \epsilon_{\alpha\beta} W_{A\alpha L} W_{A\beta L} \right]_{\mathcal{F}}, \quad (27.6.1)$$

where the superpotential  $f(\Phi)$  is a gauge-invariant cubic polynomial in the left-chiral superfields. (We are ignoring a possible  $\theta$ -term, which has no effect in perturbation theory.)

Suppose we impose an ultraviolet cut-off  $\lambda$  on the momenta circulating in loop graphs. As discussed in Section 12.4, we can find a *local* ‘Wilsonian’ effective Lagrangian density  $\mathcal{L}_\lambda$  that, with this cut-off, gives precisely the same results as the original Lagrangian density for  $S$ -matrix elements of processes at momenta below  $\lambda$ . The effective Lagrangian density has masses and coupling parameters that now depend on  $\lambda$ , and usually there will be an infinite number of coupling terms in the effective Lagrangian density, all possible terms allowed by the symmetries of the theory. But things are much simpler in supersymmetric theories. The non-renormalization theorems tell us that, as long as the cut-off preserves supersymmetry and gauge invariance, to all orders in perturbation theory the effective Lagrangian will have the structure

$$\begin{aligned} \mathcal{L}_\lambda = & \left[ \mathcal{A}_\lambda(\Phi, \Phi^\dagger, V, \mathcal{D} \dots) \right]_D + 2 \operatorname{Re} [f(\Phi)]_{\mathcal{F}} \\ & + \frac{1}{2g_\lambda^2} \operatorname{Re} \left[ \sum_{A\alpha\beta} \epsilon_{\alpha\beta} W_{A\alpha L} W_{A\beta L} \right]_{\mathcal{F}}, \end{aligned} \quad (27.6.2)$$

where  $\mathcal{A}_\lambda$  is a general Lorentz- and gauge-invariant function; ‘ $\mathcal{D} \dots$ ’ denotes terms involving superderivatives or spacetime derivatives of the preceding arguments; and  $g_\lambda$  is the *one-loop* effective gauge coupling, given by the same formula as the one-loop renormalized gauge coupling constant

$$g_\lambda^{-2} = \text{constant} - 2b \ln \lambda, \quad (27.6.3)$$

where  $b$  is the coefficient of  $g^3$  in the Gell-Mann–Low function  $\beta(g)$ , discussed in Chapter 18. This is for a simple gauge group, with a single gauge coupling, but the extension to

a direct product of simple and  $U(1)$  gauge groups is trivial. Note in particular that the effective superpotential is not only finite in the limit  $\lambda \rightarrow \infty$ , but at least in perturbation theory it contains no terms beyond those in the original superpotential, and there is no change in the coefficients of the terms it does contain.

To prove this theorem, we shall interpret this theory as a special case of one with two additional external gauge-invariant left-chiral superfields  $X$  and  $Y$ , with Lagrangian density

$$\mathcal{L}^\# = \frac{1}{2} \left[ \Phi^\dagger e^{-V} \Phi \right]_D + 2 \operatorname{Re} [Y f(\Phi)]_{\mathcal{F}} + \frac{1}{2} \operatorname{Re} \left[ X \sum_{A\alpha\beta} \epsilon_{\alpha\beta} W_{A\alpha L} W_{A\beta L} \right]_{\mathcal{F}}. \quad (27.6.4)$$

This Lagrangian density becomes equal to the original one when the scalar components  $x$  and  $y$  of  $X$  and  $Y$  are given the values  $x = 1/g^2$  and  $y = 1$ , and the spinor and auxiliary components of  $X$  and  $Y$  are set equal to zero. Since supersymmetry and gauge invariance are assumed to be preserved in the cut-off procedure, the effective Lagrangian density in the presence of these external superfields must be the sum of the  $D$ -term of a general superfield and the real part of the  $\mathcal{F}$ -term of a left-chiral superfield:

$$\begin{aligned} \mathcal{L}_\lambda^\# = & \left[ \mathcal{A}_\lambda(\Phi, \Phi^\dagger, V, X, X^\dagger, Y, Y^\dagger, \mathcal{D} \cdots) \right]_D \\ & + 2 \operatorname{Re} [\mathcal{B}_\lambda(\Phi, W_L, X, Y)]_{\mathcal{F}}, \end{aligned} \quad (27.6.5)$$

with  $\mathcal{A}_\lambda$  and  $\mathcal{B}_\lambda$  both gauge-invariant functions of the displayed arguments. We do not include any superderivatives or spacetime derivatives in the  $\mathcal{F}$ -term because, as in Section 26.3, terms involving derivatives of any of the left-chiral superfields or their adjoints may be rewritten as contributions to  $[\mathcal{A}_\lambda]_D$ . (It is true that Eq. (27.3.12) gives  $W_L$  itself in terms of two  $\mathcal{D}_{RS}$  acting on a superfield  $\exp(-2V)\mathcal{D}_L \exp(2V)$ , but this superfield is not gauge-invariant, and we are requiring that  $\mathcal{A}_\lambda$  be gauge-invariant.)

The dependence of  $\mathcal{B}_\lambda$  on  $X$  and  $Y$  is severely limited by two additional symmetries of the action obtained from the Lagrangian density (27.6.4). (Both of these symmetries are broken by non-perturbative effects, which will be considered in Chapter 29.) The first symmetry is a perturbative  $U(1)$   $R$ -symmetry, of the sort discussed in Section 26.3, for which  $\theta_L$  and  $\theta_R$  are given the  $R$  values  $+1$  and  $-1$ , the superfields  $\Phi$ ,  $V$ , and  $X$  are  $R$ -neutral, and  $Y$  has the  $R$  value  $+2$ . (Recall that  $f_{\mathcal{F}}$  is the coefficient of  $\theta_L^2$  in  $f$ , so in order for  $f_{\mathcal{F}}$  to have the  $R$  value 0,  $f$  must have the  $R$  value 2.) Because  $W_L$  is given by two  $\mathcal{D}_{RS}$  and one  $\mathcal{D}_L$  acting on  $R$ -neutral superfields, it has the  $R$  value  $+1$ . Now,  $R$  invariance requires  $\mathcal{B}_\lambda$ , like the superpotential, to have the  $R$  value  $+2$ . It cannot depend on any superfields with negative  $R$  values, such as adjoints of left-chiral superfields, because it is *holomorphic*, so the terms in  $\mathcal{B}_\lambda$  can only be of first order in  $Y$  or of second order in  $W_L$ , with coefficients depending only on the  $R$ -neutral superfields  $\Phi$  and/or  $X$ :

$$\begin{aligned} \mathcal{B}_\lambda(\Phi, W_L, X, Y) = & Y f_\lambda(\Phi, X) \\ & + \sum_{\alpha\beta AB} \epsilon_{\alpha\beta} W_{A\alpha L} W_{B\beta L} h_{\lambda AB}(\Phi, X). \end{aligned} \quad (27.6.6)$$

(Lorentz invariance requires the spinor indices on the  $W_{LS}$  to be contracted with  $\epsilon_{\alpha\beta}$ .) The other symmetry is translation of  $X$  by an imaginary numerical constant,  $X \rightarrow X + i\xi$ ,

with  $\xi$  real. This changes the Lagrangian density (27.6.4) by an amount proportional to  $\text{Im} \sum_{A\alpha\beta} W_{A\alpha L} W_{A\beta L}$ , which as we saw in Section 27.3 is a spacetime derivative, and therefore can have no effect in perturbation theory. This translation symmetry prevents  $X$  from appearing anywhere in the effective Lagrangian density (27.6.5) except where it appears in the original Lagrangian density (27.6.4). We therefore conclude that  $f_\lambda$  is independent of  $X$ , while  $h_{\lambda AB}$  consists of a  $\Phi$ -independent term proportional to  $X\delta_{AB}$ , plus a term that is independent of  $X$ . That is,

$$\begin{aligned} \mathcal{B}_\lambda(\Phi, W_L, X, Y) = & Y f_\lambda(\Phi) \\ & + \sum_{\alpha\beta AB} \epsilon_{\alpha\beta} W_{A\alpha L} W_{B\beta L} [c_\lambda \delta_{AB} X + \ell_{\lambda AB}(\Phi)] , \end{aligned} \quad (27.6.7)$$

where  $c_\lambda$  is a real cut-off-dependent constant.

The point of introducing the external auxiliary superfields  $X$  and  $Y$  is that, by giving them suitable values, we can make use of weak-coupling approximations to determine the coefficients in Eq. (27.6.7). If we set the spinor and auxiliary components of  $X$  and  $Y$  equal to zero, and take their scalar components  $x$  and  $y$  to approach infinity and zero, respectively, then the gauge coupling constant vanishes as  $1/\sqrt{x}$ , and all Yukawa and scalar couplings derived from the superpotential vanish as  $y$ . In this limit, the only graph that contributes to the term in (27.6.7) proportional to  $Y$  has a single vertex arising from the term  $2 \text{Re}[Y f(\Phi)]_{\mathcal{F}}$  in Eq. (27.6.4), so

$$f_\lambda(\Phi) = f(\Phi) . \quad (27.6.8)$$

Also, with  $Y = 0$  there is a conservation law which requires every term in  $\mathcal{L}_\lambda^\#$  to have equal numbers of  $\Phi$ s and  $\Phi^\dagger$ s, so since  $\Phi^\dagger$  cannot occur in  $\ell_{\lambda AB}$ , neither can  $\Phi$ . Gauge invariance then requires the constant  $\ell_{\lambda AB}$  to be proportional to  $\delta_{AB}$  for a simple group:

$$\ell_{\lambda AB} = \delta_{AB} L_\lambda . \quad (27.6.9)$$

Now, since gauge propagators go as  $1/x$  while pure gauge interactions go as  $x$  and scalar propagators and interactions are  $x$ -independent, with  $y = 0$  the number of powers of  $x$  in a diagram with  $V_W$  pure gauge boson vertices,  $I_W$  internal gauge boson lines, and any number of scalar-gauge boson vertices and scalar propagators is

$$N_x = V_W - I_W . \quad (27.6.10)$$

The number of loops is given by

$$L = I_W + I_\Phi - V_W - V_\Phi + 1 , \quad (27.6.11)$$

where  $I_\Phi$  is the number of internal  $\Phi$  lines and  $V_\Phi$  is the number of  $\Phi$ - $V$  interaction vertices. All the  $\Phi$ - $V$  vertices have two  $\Phi$  lines attached, so with no external  $\Phi$  lines  $I_\Phi$  and  $V_\Phi$  are equal, and therefore cancel in Eq. (27.6.11), so that Eq. (27.6.10) may be written

$$N_x = 1 - L . \quad (27.6.12)$$

Thus the coefficient  $c_\lambda$  of  $X$  in Eq. (27.6.7) is correctly given by the tree approximation and is therefore what it was in the original Lagrangian, simply  $c_\lambda = 1$ , while the coefficient  $L_\lambda$  of the  $X$ -independent term is given by one-loop diagrams only. Putting this all together, we have

$$\begin{aligned} \mathcal{L}_\lambda^\# = & \left[ \mathcal{A}_\lambda(\Phi, \Phi^\dagger, V, X, X^\dagger, Y, Y^\dagger, \mathcal{D} \dots) \right]_D + 2 \operatorname{Re} [Y f(\Phi)]_{\mathcal{F}} \\ & + \frac{1}{2} \operatorname{Re} \left[ (X + L_\lambda) \sum_{A\alpha\beta} \epsilon_{\alpha\beta} W_{A\alpha L} W_{A\beta L} \right]_{\mathcal{F}}, \end{aligned} \quad (27.6.13)$$

where  $L_\lambda$  is the one-loop contribution. Setting  $Y = 1$  and  $X = 1/g^2$  then gives Eq. (27.6.2), with  $g_\lambda^{-2} = g^{-2} + L_\lambda$ . As shown in Section 18.3, the leading order contribution to  $\lambda dg_\lambda/d\lambda$  is the same function of  $g_\lambda$  whatever renormalization scheme is used to define this coupling, so to one-loop order we must have

$$\lambda \frac{dg_\lambda}{d\lambda} = b g_\lambda^3, \quad (27.6.14)$$

where  $b$  is the same coefficient of  $g^3$  as in the renormalization group equation of Gell-Mann and Low. The solution is Eq. (27.6.3), completing the proof.

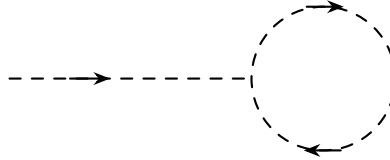
In theories with a  $U(1)$  gauge superfield  $V_1$ , the Lagrangian may contain a Fayet–Iliopoulos term (27.2.7):

$$\mathcal{L}_{\text{FI}} = \xi [V_1]_D. \quad (27.6.15)$$

It is easy to see that the coefficient  $\xi$  of such a term is not renormalized [7]. If the corresponding coefficient  $\xi_\lambda$  in the Wilsonian Lagrangian density did depend on the gauge couplings or the couplings in the superpotential, then when we replace the original Lagrangian (27.6.1) with a Lagrangian (27.6.4) involving the external superfields  $X$  and  $Y$ , this term in the Wilsonian Lagrangian would be required by supersymmetry to take the form

$$\mathcal{L}_{\text{FI}\lambda}^\# = [\xi_\lambda(X, Y, X^*, Y^*) V_1]_D, \quad (27.6.16)$$

with  $\xi_\lambda$  a function with a non-trivial dependence on  $X$  and/or  $Y$  and/or their adjoints. But such a term would not be gauge-invariant, because, according to Eq. (27.2.18), a gauge transformation shifts  $V_1$  by a chiral superfield  $i(\Omega - \Omega^*)/2$ , and although the  $D$ -term of a chiral superfield vanishes, the product of  $i(\Omega - \Omega^*)/2$  and  $\xi_\lambda$  is not chiral for general gauge transformations if  $\xi_\lambda$  has any dependence on other superfields. There actually are diagrams that make contributions to  $\xi_\lambda$  that are independent



**Figure 27.1.** A one-loop diagram that could be quadratically divergent in theories with supersymmetry broken by trilinear couplings among scalar fields and their adjoints. The lines all represent complex scalar fields.

of all coupling constants. For the Lagrangian (27.6.1) there are no factors of the gauge coupling  $g$  at vertices at which the gauge superfield interacts with chiral matter, but instead a factor  $g^{-2}$  for each gauge propagator, so a graph with no internal gauge lines and no self-couplings of the chiral superfield will have no dependence on coupling constants. The only such graphs that contribute to  $\xi_\lambda$  are those in which a single external gauge line is attached to a chiral loop. (See Figure 27.1.) The contribution of all such graphs is proportional to the sum of the gauge couplings of all chiral superfields—that is, to the trace of the  $U(1)$  generator. But as discussed in Section 22.4, this trace must vanish (if the  $U(1)$  symmetry is unbroken) in order to avoid gravitational anomalies that violate the conservation of the  $U(1)$  current.

The most important application of these theorems is a corollary, which tells us that if there is no Fayet–Iliopoulos term and if the superpotential  $f(\Phi)$  allows solutions of the equations  $\partial f(\phi)/\partial\phi_n = 0$ , then supersymmetry is not broken in any finite order of perturbation theory.

To test this we must examine Lorentz-invariant field configurations, in which the  $\Phi_n$  have only constant scalar components  $\phi_n$  and constant auxiliary components  $\mathcal{F}_n$ , while (in Wess–Zumino gauge) the coefficients  $V_A$  of the gauge generators  $t_A$  in the matrix gauge superfield  $V$  have only auxiliary components  $D_A$ . Supersymmetry is unbroken if there are values of  $\phi_n$  for which  $\mathcal{L}_\lambda$  has no terms of first order in  $\mathcal{F}_n$  or  $D_A$ , in which case there is sure to be an equilibrium solution with  $\mathcal{F}_n = D_A = 0$ . (In Section 29.2 we will see that this is the sufficient as well as the necessary condition for supersymmetry to be unbroken.) In the absence of Fayet–Iliopoulos terms, this will be the case if for all  $A$

$$\sum_{nm} \frac{\partial K_\lambda(\phi, \phi^*)}{\partial \phi_n^*} (t_A)_{mn} \phi_m^* = 0 \quad (27.6.17)$$

and for all  $n$

$$\frac{\partial f(\phi)}{\partial \phi_n} = 0, \quad (27.6.18)$$

where the effective Kähler potential  $K_\lambda(\phi, \phi^*)$  is

$$K_\lambda(\phi, \phi^*) = \mathcal{A}_\lambda(\phi, \phi^*, 0, 0 \cdots), \quad (27.6.19)$$

with  $\mathcal{A}_\lambda(\phi, \phi^*, 0, 0 \cdots)$  obtained from  $\mathcal{A}_\lambda$  by setting the gauge superfield and all superderivatives equal to zero. (With superderivatives required to vanish by Lorentz invariance, the only dependence of  $\mathcal{A}_\lambda$  on  $V$  is a factor  $\exp(-V)$  following every factor  $\Phi^\dagger$ .)

We now use a trick that we have already employed in Section 27.4. If there is any solution  $\phi^{(0)}$  of Eq. (27.6.18), then the gauge symmetry tells us that there is a continuum of such solutions, with  $\phi_n$  replaced with

$$\phi_n(z) = \left[ \exp \left( i \sum_A t_A z_A \right) \right]_{nm} \phi_m^{(0)}, \quad (27.6.20)$$

where (since  $f$  depends only on  $\phi$ , not  $\phi^*$ ) the  $z_A$  are an arbitrary set of *complex* parameters. If  $K_\lambda(\phi, \phi^*)$  has a stationary point anywhere on the surface  $\phi = \phi(z)$ , then at that point

$$0 = \sum_{nmA} \frac{\partial K_\lambda(\phi, \phi^*)}{\partial \phi_n} (t_A)_{nm} \phi_m \delta z_A - \sum_{nmA} \frac{\partial K_\lambda(\phi, \phi^*)}{\partial \phi_n^*} (t_A)_{mn} \phi_m^* \delta z_A^* . \quad (27.6.21)$$

Since this must be satisfied for all infinitesimal complex  $\delta z_A$ , the coefficients of both  $\delta z_A$  and  $\delta z_A^*$  must both vanish, and therefore Eq. (27.6.17) as well as Eq. (27.6.18) is satisfied at this point. Thus the existence of a stationary point of  $K_\lambda(\phi, \phi^*)$  on the surface  $\phi = \phi(z)$  would imply that supersymmetry is unbroken to all orders of perturbation theory.

The zeroth-order Kähler potential  $(\phi^\dagger \phi)$  is bounded below and goes to infinity as  $\phi \rightarrow \infty$ , so it certainly has a minimum on the surface  $\phi = \phi(z)$ , where of course it is stationary. If there were no flat directions in which  $K_\lambda$  is constant at this minimum then any sufficiently small perturbation to the Kähler potential might shift the minimum, but would not destroy it. At the minimum of the Kähler potential on the surface  $\phi = \phi(z)$  there are flat directions: ordinary global gauge transformations  $\delta \phi = i \sum_A \delta z_A t_A \phi$  with  $z_A$  real. But these are also flat directions for the perturbation  $K_\lambda(\phi, \phi^*) - (\phi^\dagger \phi)$ , so there is still a local minimum of  $K_\lambda$  on the surface  $\phi = \phi(z)$  for any perturbation in at least a finite range, and thus to all orders in whatever couplings appear in  $K_\lambda(\phi, \phi^*)$ . As we have seen, this is a set of scalar field values at which  $\mathcal{F}_n = 0$  and  $D_A = 0$  for all  $n$  and  $A$ , which means that supersymmetry is unbroken.

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These results may be extended to non-renormalizable theories [3]. In such theories the first term  $[\Phi^\dagger e^{-V} \Phi]_D$  in Eq. (27.6.1) is replaced with the  $D$ -term of an arbitrary real gauge-invariant scalar function of  $\Phi^\dagger$ ,  $\Phi$ ,  $V$ , and their superderivatives and spacetime derivatives, while the second and third terms in Eq. (27.6.1) are replaced with the  $\mathcal{F}$ -term of an arbitrary globally gauge-invariant scalar function  $f(\Phi, W)$  of  $\Phi_n$  and  $W_\alpha$ . It has been shown that, to all orders of perturbation theory, the function  $f_\lambda(\Phi, W)$  appearing in the  $\mathcal{F}$ -term of the Wilsonian Lagrangian is the same as  $f(\Phi, W)$ , except for the one-loop renormalization of the term quadratic in  $W$ .